Normal Equation in Linear Regression

- The **Normal Equation** is an analytical, closed-form approach to computing the parameter vector $\boldsymbol{\theta}$ that minimizes the Mean Squared Error (MSE) cost function in Linear Regression. Unlike iterative optimization algorithms (e.g., Gradient Descent), the Normal Equation computes the exact optimal parameters directly in a single mathematical step without requiring hyperparameter tuning such as selecting a learning rate α.
- In ordinary least squares (OLS) linear regression, given a design matrix $X \in \mathbb{R}^{m\times(n+1)}$ (including the intercept column of 1s) and a target vector $\mathbf{y} \in \mathbb{R}^m$, the model predictions are given by $\hat{\mathbf{y}} = X\boldsymbol{\theta}$.

- The cost function $J(\boldsymbol{\theta})$ represents the sum of squared residual errors:

$$J(\boldsymbol{\theta}) = \frac{1}{2m}(\mathbf{y} - X\boldsymbol{\theta})^T(\mathbf{y} - X\boldsymbol{\theta})$$

To find the minimum of $J(\theta)$, take the partial derivative with respect to θ and set it to zero:

1. **Expand the cost function:**

$$J(\boldsymbol{\theta}) \propto \mathbf{y}^T\mathbf{y} - \boldsymbol{\theta}^T X^T\mathbf{y} - \mathbf{y}^T X\boldsymbol{\theta} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}$$

   Since $\boldsymbol{\theta}^T X^T\mathbf{y}$ is a scalar, it equals its transpose $\mathbf{y}^T X\boldsymbol{\theta}$:

$$J(\boldsymbol{\theta}) \propto \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\theta}^T X^T\mathbf{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}$$

2. **Compute the gradient with respect to $\boldsymbol{\theta}$:**

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = -2X^T\mathbf{y} + 2X^T X\boldsymbol{\theta}$$

3. **Set the gradient to zero to solve for $\boldsymbol{\theta}$:**

$$-2X^T\mathbf{y} + 2X^T X\boldsymbol{\theta} = \mathbf{0}$$

$$X^T X\boldsymbol{\theta} = X^T\mathbf{y}$$

4. **Isolate $\boldsymbol{\theta}$ (assuming $X^T X$ is invertible):**

$$\boldsymbol{\theta} = (X^T X)^{-1} X^T \mathbf{y}$$

- **Computing Optimal Parameters:**

To apply this closed-form solution:

1. Construct the design matrix X by prepending a column of 1s to the feature matrix.
2. Compute the matrix product $X^T X$.
3. Invert $(X^T X)$ and multiply by $X^T \mathbf{y}$ to yield the parameter vector $\boldsymbol{\theta} = [\theta_0, \theta_1, \dots, \theta_n]^T$.

- **Normal Equation vs. Gradient Descent:**

Characteristic	Normal Equation	Gradient Descent
Method Type	Analytical / Closed-form	Iterative optimization
Learning Rate (α)	No choice required	Must be carefully tuned
Computational Complexity	$\mathcal{O}(n^3)$ due to matrix inversion	$\mathcal{O}(k \cdot n \cdot m)$ for k iterations
Feature Scaling	Not required	Essential for fast convergence
Performance with Large n	Slow for $n >$ $10{,}000$features	Scales efficiently to millions of features
Invertibility Dependency	Requires $X^T X$ to be invertible	Works regardless of matrix invertibility